

JAVA PROGRAMMING LABORATORY

Experiment No. 05

Date	17/09/2026
Title	Digital Wallet / UPI Payment System using OOP and Custom Exceptions

Objective / Aim

To write and execute a Java program that demonstrates object-oriented programming concepts, interfaces, encapsulation, custom exception handling, and a menu-driven digital wallet with UPI payments.

Hardware Requirements

Standard PC/Laptop, Core i3/i5 or equivalent processor, 4 GB/8 GB RAM.

Algorithm / Logic Description

1. Import the Scanner class for keyboard input.
2. Define custom checked exceptions: InvalidUPIException, InvalidAmountException, and InsufficientBalanceException.
3. Define the PaymentService interface with pay() and checkBalance() methods.
4. Create the Wallet class with private user details and balance to demonstrate encapsulation.
5. Implement methods to add money, get balance, deduct money, and display wallet details.
6. Create UPIPayment that implements PaymentService and performs UPI ID, amount, and balance validation.
7. Throw the appropriate custom exception when validation fails.
8. In main(), accept wallet details and create Wallet and UPIPayment objects.
9. Display a menu for adding money, making UPI payments, checking balance, displaying details, and exiting.
10. Handle exceptions using try-catch-finally and display suitable transaction messages.

UML Class Diagram

<<interface>> PaymentService
+ pay(upild, amount)
+ checkBalance()

Wallet
- userName : String
- mobileNumber : String
- upild : String
- balance : double
+ Wallet(...)
+ addMoney(amount)
+ getBalance()
+ deductMoney(amount)
+ displayWalletDetails()

implements

UPIPayment
- wallet : Wallet
+ UPIPayment(wallet)
+ pay(upild, amount)
+ checkBalance()

uses →

Custom Exceptions
InvalidUPIException
InvalidAmountException
InsufficientBalanceException

Relationships: UPIPayment implements PaymentService and has a Wallet reference. The custom exception classes inherit from Java's Exception class.

Source Code

```
package observation.exp05;
import java.util.Scanner;

// ===== CUSTOM EXCEPTIONS =====

class InsufficientBalanceException extends Exception {
    public InsufficientBalanceException(String message) {
        super(message);
    }
}

class InvalidUPIException extends Exception {
    public InvalidUPIException(String message) {
        super(message);
    }
}

class InvalidAmountException extends Exception {
    public InvalidAmountException(String message) {
        super(message);
    }
}

// ===== INTERFACE =====

interface PaymentService {
    void pay(String upiId, double amount)
        throws InvalidUPIException,
               InvalidAmountException,
               InsufficientBalanceException;

    void checkBalance();
}

// ===== WALLET CLASS =====
```

```

class Wallet {
    private String userName;
    private String mobileNumber;
    private String upiId;
    private double balance;

    public Wallet(String userName, String mobileNumber,
                  String upiId, double balance) {
        this.userName = userName;
        this.mobileNumber = mobileNumber;
        this.upiId = upiId;
        this.balance = balance;
    }

    public void addMoney(double amount)
        throws InvalidAmountException {
        if (amount <= 0) {
            throw new InvalidAmountException(
                "Amount must be greater than zero."
            );
        }

        balance = balance + amount;

        System.out.println(
            "■" + amount + " added successfully."
        );
    }

    public double getBalance() {
        return balance;
    }

    public void deductMoney(double amount) {
        balance = balance - amount;
    }
}

```

Source Code (continued)

```
public void displayWalletDetails() {
    System.out.println("\n===== WALLET DETAILS =====");
    System.out.println("User Name      : " + userName);
    System.out.println("Mobile Number : " + mobileNumber);
    System.out.println("UPI ID       : " + upiId);
    System.out.println("Balance      : ■" + balance);
}

// ===== UPI PAYMENT CLASS =====

class UPIPayment implements PaymentService {
    private Wallet wallet;

    public UPIPayment(Wallet wallet) {
        this.wallet = wallet;
    }

    public void pay(String upiId, double amount)
        throws InvalidUPIException,
               InvalidAmountException,
               InsufficientBalanceException {

        if (upiId == null ||
            !upiId.contains("@") ||
            upiId.startsWith("@") ||
            upiId.endsWith("@")) {
            throw new InvalidUPIException(
                "Invalid UPI ID."
            );
        }

        if (amount <= 0) {
            throw new InvalidAmountException(
                "Payment amount must be greater than zero."
            );
        }

        if (amount > wallet.getBalance()) {
            throw new InsufficientBalanceException(
                "Insufficient wallet balance."
            );
        }

        wallet.deductMoney(amount);

        System.out.println("\n===== PAYMENT SUCCESSFUL =====");
        System.out.println("Paid To : " + upiId);
        System.out.println("Amount  : ■" + amount);
    }

    public void checkBalance() {
        System.out.println(
            "Available Balance: ■" +
            wallet.getBalance()
        );
    }
}

// ===== MAIN CLASS =====

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("===== CREATE WALLET =====");

        System.out.print("Enter your name: ");
        String name = sc.nextLine();

        System.out.print("Enter mobile number: ");
```

Source Code (continued)

```
String mobile = sc.nextLine();

System.out.print("Enter your UPI ID: ");
String myUpi = sc.nextLine();

System.out.print("Enter initial balance: ");
double balance = sc.nextDouble();

Wallet wallet = new Wallet(
    name, mobile, myUpi, balance
);

UPIPayment payment = new UPIPayment(wallet);

int choice;

while (true) {
    System.out.println("\n=====");
    System.out.println("        DIGITAL WALLET");
    System.out.println("=====");
    System.out.println("1. Add Money");
    System.out.println("2. Make UPI Payment");
    System.out.println("3. Check Balance");
    System.out.println("4. Display Wallet Details");
    System.out.println("5. Exit");
    System.out.println("=====");

    System.out.print("Enter your choice: ");
    choice = sc.nextInt();

    switch (choice) {
        case 1:
            System.out.print("Enter amount to add: ");
            double addAmount = sc.nextDouble();

            try {
                wallet.addMoney(addAmount);
            }
            catch (InvalidAmountException e) {
                System.out.println(
                    "Transaction Failed: " +
                    e.getMessage()
                );
            }
            break;

        case 2:
            sc.nextLine();

            System.out.print("Enter receiver UPI ID: ");
            String receiverUpi = sc.nextLine();

            System.out.print("Enter payment amount: ");
            double paymentAmount = sc.nextDouble();

            try {
                payment.pay(receiverUpi, paymentAmount);
            }
            catch (InvalidUPIException e) {
                System.out.println(
                    "Transaction Failed: " +
                    e.getMessage()
                );
            }
            catch (InvalidAmountException e) {
                System.out.println(
                    "Transaction Failed: " +
                    e.getMessage()
                );
            }
            catch (InsufficientBalanceException e) {
                System.out.println(
```

Source Code (continued)

```
        "Transaction Failed: " +
        e.getMessage()
    );
}
finally {
    System.out.println(
        "Transaction processing completed."
    );
}
break;

case 3:
    payment.checkBalance();
    break;

case 4:
    wallet.displayWalletDetails();
    break;

case 5:
    System.out.println(
        "\nThank you for using Digital Wallet!"
    );
    sc.close();
    return;

default:
    System.out.println(
        "Invalid choice! Please enter 1 to 5."
    );
}
}
}
}
```

Compilation & Execution Commands

```
cd observation/exp05
javac Main.java
cd ../../
java observation.exp05.Main
```

Because the supplied program declares package `observation.exp05` and the public class is `Main`, the source file should be saved as `Main.java` inside the `observation/exp05` directory. The commands above compile and run the program from the project root.

Sample Output Screen

```
===== CREATE WALLET =====
Enter your name: Tharun
Enter mobile number: 9876543210
Enter your UPI ID: tharun@upi
Enter initial balance: 1000

=====
                DIGITAL WALLET
=====
1. Add Money
2. Make UPI Payment
3. Check Balance
4. Display Wallet Details
5. Exit
=====
Enter your choice: 2
Enter receiver UPI ID: friend@upi
Enter payment amount: 250

===== PAYMENT SUCCESSFUL =====
Paid To : friend@upi
Amount : ■250.0
Transaction processing completed.

=====
                DIGITAL WALLET
=====
Enter your choice: 3
Available Balance: ■750.0

Enter your choice: 5

Thank you for using Digital Wallet!
```

Result

The Digital Wallet / UPI Payment System was implemented successfully in Java using an interface, encapsulation, custom checked exceptions, exception handling, constructors, and a menu-driven program.